

 SPEAKER 1 — CHEW WEI JIAN

Problem Significance, Dataset & The First Finding (~52 sec)

[VISUAL: Title card + Predictive_Limits_of_Dating_Data.png]

Hi everyone. We are Group 3, OCC6, from the Faculty of Computer Science and Information Technology, Universiti Malaya. Our project is called "Tying the Data Knot" — a machine learning investigation into whether we can predict meaningful human connections on a dating app.

[VISUAL: 01_match_outcome_distribution.png]

Our dataset contains 50,000 user profiles across 25 features — demographics, behavioural metrics like swipe ratio, message count, and mutual matches. We binarised 10 original outcome categories into two classes: positive outcomes — mutual matches, instant matches, dates, and relationships formed— versus negative outcomes like ghosting, blocking, and no action.

[VISUAL: 05_numerical_features_by_match_outcome.png]

The first and most important finding from our comprehensive EDA: despite a realistic class imbalance of 39.7% positive to 60.3% negative, feature distributions across both classes are nearly identical. Swipe ratios, app usage, message counts — essentially flat. No single feature separates matches from non-matches.

This is not a failure. It is a scientific result — and it shaped every design decision that follows.

 SPEAKER 2 — KU JIAN CHENG

Preprocessing, Causal Discovery & Feature Engineering (~67 sec)

[VISUAL: assets/pipeline_diagram.png]

Thank you Wei Jian. Before training a single model, we built a production-grade preprocessing pipeline driven entirely by what the data told us.

For encoding: ordinal encoding for ranked features like income and education, one-hot encoding for nominal categories, and multi-hot encoding for 49 interest tags into a binary matrix. Together with three engineered interaction features, this expanded our 25 raw columns to 116 input features.

[VISUAL: 09_causal_dag.png]

Beyond encoding, we applied the PC Algorithm with a Kernel Conditional Independence test to map the causal DAG of user behaviour. This revealed `mutual_matches` is a collider variable — so using it as a predictor introduces structural bias. Double Machine

Learning then isolated the causal effect of profile photos: an ATE of 0.0104 with $p = 0.0322$ — a statistically significant but inherently weak signal.

[VISUAL: 11_ood_anomaly_score_distribution.png]

RobustScaler was chosen for skewed activity metrics. An Isolation Forest OOD guardrail rejects anomalous profiles at inference. Feature selection combined ANOVA F-Score, Mutual Information, and Boruta in parallel — pruning 116 features down to 67.

[VISUAL: 16_train_test_class_distribution.png]

A stratified 80/20 split with SMOTE applied only on training data ensured no synthetic samples leak into the test set.

 SPEAKER 3 — NG JIN RU

16-Architecture Benchmark & Champion Selection (~70 sec)

[VISUAL: 18_model_performance_comparison.png]

Thank you Jian Cheng. With our data pipeline locked, we moved to model training. We built and evaluated 16 architectures in total — 13 baseline classifiers spanning every major paradigm, plus three advanced architectures: a Graph Attention Network, a SCARF self-supervised learner, and a TabPFN Zero-Shot Transformer.

[VISUAL: 19_confusion_matrices.png + 20_roc_curves.png]

All models cluster around ROC-AUC of 0.51 — barely above random chance. The Friedman-Nemenyi test confirmed differences are statistically significant ($p = 0.0344$), proving our baseline metrics are not due to random variance, and FLAML and PyCaret AutoML validation hit the exact same 60.3% ceiling — proving we extracted 100% of the available signal.

[VISUAL: 29_all_models_roc_auc_ranking.png]

Champion selection required careful justification. SVM Bagging scored the highest raw ROC-AUC of 0.5143 — but with 0% F1-Score, it had collapsed into a majority-class predictor. Scientifically meaningless. LightGBM scored ROC-AUC 0.5112 with F1 of 18.57% — one of the few models successfully maintaining balanced positive-class discrimination without fully collapsing global accuracy. Its speed enabled 1,000-trial Optuna tuning, and its native SHAP TreeExplainer compatibility powers our entire downstream pipeline: calibration, DiCE recourse, Knowledge Distillation, and Causal Uplift. A model that breaks any of these collapses the system.

 SPEAKER 4 — ANG YING EN

Tuning, Evaluation & Advanced Validation (~62 sec)

[VISUAL: 26_baseline_vs_tuned_comparison.png]

Thank you Jin Ru. With our champion selected, here is how we optimised and validated it.

We tuned LightGBM with RandomizedSearchCV and GPU-accelerated Optuna — using Matthews Correlation Coefficient as the objective, because MCC is the only metric that scores high when all four confusion matrix quadrants perform well simultaneously. This prevents the optimiser from gaming accuracy on the majority class.

[VISUAL: 22_learning_curves.png]

Learning curves showed tight train-validation convergence — definitive proof the ceiling is a data limitation, not a modelling one. We calibrated LightGBM with Isotonic Regression on a held-out split, ensuring probabilities reflect true empirical frequencies.

[VISUAL: 32_shap_interaction_matrix +
36_probability_calibration_reliability_diagram.png]

For interpretability, we computed SHAP interaction values and Friedman's H-Statistic to map pairwise feature synergies. We also ran a demographic parity audit — confirming our model does not discriminate by gender or orientation. For robustness, Conformal Prediction provides provable uncertainty coverage, and FGSM adversarial testing verified graceful degradation. For deployment, Knowledge Distillation compressed our champion into a lightweight model, and Microsoft DiCE generates actionable profile recommendations — turning predictions into user agency.

 SPEAKER 5 — CHAANG WAI CHIU

Deployment, Conclusion & Future Work (~53 sec)

[VISUAL: 38_causal_uplift_targeting_segments.png → streamlit_overview.png → streamlit_causal_uplift.png]

We also built a T-Learner Causal Uplift model to segment users by individual treatment effect, and deployed SwipelQ V2 — a Streamlit dashboard with nine live sandboxes covering every pipeline component.

[VISUAL: Decoding_Romantic_Success_Research.png]

To conclude: we evaluated 16 architectures, disqualified majority-class predictors, and selected LightGBM as our champion — validating the 60.3% ceiling with AutoML, statistical testing, and learning curves. That limit is a property of the data, not our engineering.

For future work, our report proposes five directions: diffusion-based synthetic augmentation, tabular transfer learning, graph neural networks for bipartite

matchmaking, Bayesian neural networks for calibrated intervals, and active learning with human-in-the-loop surveys.

Our scientific finding is clear: with the features available, love cannot be scientifically predicted. But the rigour of that finding — the 17-section pipeline we built to prove it honestly — that is the real contribution of this project. Thank you.